Status	Finished
Started	Saturday, 10 October 2026, 6:09 PM
Completed	Saturday, 10 October 2026, 6:15 PM
Duration	6 mins 15 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n/2
b. n-1
c. log n
d. n ✓

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. In the left sublist
c. At the beginning of the list
d. In the middle of the list

Question **3**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search continues until the end of the list
c. The search starts over from the beginning
d. The search backtracks to find duplicate elements

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Dividing the list in half repeatedly
c. Skipping every second element
d. Checking each element sequentially ✓

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list
b. It compares elements sequentially
c. It can be applied only if the list is sorted ✓
d. It works on unsorted lists

Question **6**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Deleting elements from a list
c. Inserting elements into a list
d. Determining whether an element is present in a list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By starting from the first element
c. By comparing each element sequentially
d. By dividing the list length by two ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. An error is raised
d. The search is considered unsuccessful ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)
b. O(log n) ✓
c. O(n log n)
d. O👎

Question **10**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. When the item is not the array at all
c. Item is the last element in the array or item is not there at all
d. When the item is somewhere in the middle of the array ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99 ✓
b. 89 and 94
c. 90 and 100
d. 94 and 99

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The search continues in the left or right sublist ✓
d. The list is sorted

Question **13**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search has a lower time complexity on large, sorted lists ✓
c. Binary search does not require dividing the list
d. Binary search can find multiple instances of the target element

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O👎
c. O(1) ✓
d. O(n log n)

Question **15**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. log n
c. n ✓
d. n/2